

Sumedh Kothari

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EDUCATION

Georgia Institute of Technology

B.S. in College of Engineering

Expected May 2028

GPA: 3.82

- **Coursework:** Computer Organization and Programming; Computing for Engineers; Advanced Data Structures, Embedded Systems, and Networking; Ordinary Differential Equations
- **Honors:** Symposium of Rising Scholars Presenter, Eagle Scout

EXPERIENCE

Georgia Tech Renewable Bioproducts Institute | Tong Laboratory

Research Intern

Atlanta, GA

August 2025 – Present

- Used Python, NumPy/Pandas libraries to apply Bayesian statistical modeling to optimize lignin valorization reaction
- Utilized NMR, SEM, and HPLC modeling data to characterize value-added chemicals from lignin breakdown
- Completed comprehensive literature reviews on aromatic compound synthesis, 3D printing custom GO structures, and optimal catalyst design while collaborating with research team to write manuscript for publication

HyTech Racing | Georgia Tech

Embedded Systems Engineer

Atlanta, GA

February 2026 - Present

- Designed and implemented a real-time lap tracking module in modern C++ for Georgia Tech's Formula SAE Racing team, leveraging Protobuf for low-latency telemetry serialization and inter-process communication across vehicle subsystems
- Developing and deploying a YOLO-based object detection pipeline for driverless perception, enabling real-time cone detection and localization through C++ inference integration into the autonomous control stack

ChemE Cube | Georgia Tech

Team Lead – Mechanical

Atlanta, GA

August 2025 – Present

- Leading the mechanical development of a 1 ft³ direct air capture reactor by managing a 5-member subteam and overseeing end-to-end CAD, prototyping, and systems integration, delivering a fully fabricated and test-ready prototype
- Designing airflow distribution and sorbent containment subsystems using parametric CAD, 3D printing, and CNC machining, achieving 75% reduction of ambient CO₂ in bench-scale validation for AIChE competition (Nov 2026)

PROJECTS

IEEE Robotech 2026 Hackathon - Autonomous Track | Georgia Tech

Mechanical/Software Engineer

Atlanta, GA

January 2026

- Designed and CAD-modeled a fully autonomous debris-clearing robot by engineering the chassis, actuation mechanisms, and beacon-deployment system, enabling end-to-end autonomous operation without human intervention
- Trained, developed, and deployed a YOLO-based computer vision model in PyTorch by integrating it into the onboard perception pipeline, achieving 95% debris detection accuracy and enabling real-time hazard identifications
- Integrated perception, navigation, and mechanical actuation systems by linking CV outputs to motion planning and debris-removal hardware, allowing the robot to autonomously clear obstacles to certify landing safety

EcoFlow - Sustainability Calculator | Python, FastAPI, Next.js, Redis, PostgreSQL

Developer

Atlanta, GA

January 2026

- Built and deployed a FastAPI (Python) + Next.js (TypeScript) full-stack web app that processes any Amazon ASIN across 5 lifecycle stages, automating cradle-to-grave LCA calculations and reducing analysis time from hours to seconds via AI-driven product parsing and structured emissions modeling
- Engineered asynchronous REST APIs with Redis caching (24-hr TTL) and PostgreSQL persistence, containerizing services with Docker to achieve <1s repeat-query latency while improving scalability and cost efficiency

AiApply - AI Job Applier | Python, Selenium WebDriver, HTML, JavaScript

Developer

Atlanta, GA

October 2025

- Developed full-stack AI-powered internship application platform to automate end-to-end job search workflows, including resume optimization, cover letter generation, and programmatic application submission on Handshake
- Implemented Selenium-based automation to extract recruiter profiles from filtered company searches and deliver personalized direct messages at scale, increasing outreach efficiency and response rates
- Engineered web scraping algorithms to identify target companies based on location and industry parameters, then automated tailored email outreach highlighting candidate skillsets parsed from resume data

Sports Betting Arbitrage Calculator | Python, REST APIs

Developer

Atlanta, GA

November 2025

- Developed application that aggregates live odds from multiple sportsbooks via API, normalizes data, and identifies risk-free arbitrage opportunities across multiple market types (H2H, Spread, Total) with up to 50% daily ROI.
- Built a robust, production-ready pipeline with Dockerization, session management, error handling, and logging, enabling scalable, real-time analysis across dozens of active sports markets.

SKILLS

- **Technical:** Python, C++, FastAPI, Next.js, Redis, PostgreSQL, Data analysis (Excel, MATLAB), CAD (SolidWorks)
- **Communication:** Technical Writing, Multilingual (English, Hindi, Spanish)